

GRYTVIKEN, South Georgia and the South Sandwich Islands

WMO: 889030

Lat: 54.284S

Lon: 36.496W

Elev: 3

StdP: 101.29

Time Zone: -2.00 (W02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -8.6	(c) -7.3	(d) -15.2	(e) 1.0	(f) -5.3	(g) -13.4	(h) 1.2	(i) -5.0	(j) 14.9	(k) 0.3	(l) 13.5	(m) 2.3	(n) 2.5	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 6.8	(c) 15.8	(d) 10.0	(e) 14.0	(f) 9.1	(g) 12.3	(h) 7.8	(i) 10.7	(j) 14.6	(k) 9.4	(l) 12.9	(m) 8.2	(n) 11.5	(o) 8.1	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	6.7	12.5	6.8	6.1	11.2	5.8	5.7	9.8	31.3	14.8	28.2	12.9	25.4	11.5	16.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.3	(b) 12.6	(c) 11.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	6.0	6.8	5.1	2.9	1.2	-0.3	-1.1	-1.2	0.8	2.6	4.2	5.4
	DBStd	3.93	2.31	2.80	2.44	2.77	3.53	3.02	3.33	3.16	3.01	2.72	2.48	2.53
	HDD10.0	2687	126	98	153	213	275	310	344	346	275	228	174	144
	HDD18.3	5713	382	323	409	463	532	560	602	604	525	486	424	401
	CDD10.0	17	3	8	2	0	1	0	0	0	0	0	1	2
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	4.6	4.7	5.3	4.9	4.3	4.3	4.1	4.3	4.6	4.9	4.6	4.8	4.9
(15) Precipitation	PrecAvg	1647	120	129	174	161	167	146	144	150	139	123	104	91
	PrecMax	1971	192	172	248	257	251	203	238	241	310	212	153	131
	PrecMin	1288	69	61	135	104	112	93	63	103	23	65	49	51
	PrecStd	199	33	29	30	32	29	28	34	32	45	29	20	22
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	19.0	16.6	14.1	14.0	10.5	12.0	9.1	10.4	13.6	15.0	16.6
		MCWB	11.2	12.8	10.2	9.0	9.7	5.0	6.7	5.3	5.5	8.6	8.4	9.7
	2%	DB	13.9	16.0	13.6	11.9	11.1	7.7	7.4	7.1	8.3	11.0	12.2	13.3
		MCWB	8.5	10.5	9.1	7.7	7.1	3.6	3.6	3.5	4.4	5.4	8.0	8.0
	5%	DB	11.7	14.1	11.3	9.1	8.4	5.4	5.5	5.5	6.8	9.0	10.1	11.6
		MCWB	7.0	9.8	7.7	5.5	5.0	2.4	2.5	2.4	3.1	4.9	5.9	7.2
	10%	DB	9.9	11.8	9.5	7.0	6.1	3.9	3.9	4.0	5.4	7.3	8.7	10.1
		MCWB	6.1	7.9	6.4	4.6	3.2	1.3	1.3	1.4	2.4	3.8	4.9	6.0
	0.4%	WB	12.2	13.9	11.7	10.1	9.5	6.5	6.1	5.7	5.8	8.3	10.0	9.7
		MCDB	17.4	18.8	16.1	13.6	13.4	9.8	8.8	8.5	9.1	11.6	13.1	13.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	9.1	11.3	9.3	8.0	7.2	3.9	4.0	4.1	4.7	6.0	8.1	8.4
		MCDB	12.7	15.4	12.3	11.7	10.3	6.7	6.8	6.3	7.8	9.0	11.1	11.9
	5%	WB	7.4	9.8	8.0	6.5	5.1	2.6	2.7	2.6	3.6	4.9	6.5	7.2
		MCDB	10.6	13.3	10.8	9.1	8.0	5.2	5.1	4.9	6.3	7.9	9.5	10.8
	10%	WB	6.5	8.0	6.7	4.9	3.4	1.6	1.5	1.6	2.6	3.9	5.4	6.2
		MCDB	9.3	11.5	9.1	6.7	5.8	3.5	3.5	3.6	4.9	6.6	7.9	9.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	6.4	5.9	4.9	4.5	4.6	5.1	5.5	6.7	6.2	6.8
		MCDBR	9.5	10.7	10.4	9.4	8.1	7.7	6.8	7.1	7.7	9.6	9.3	9.8
		MCWBR	5.7	6.8	7.0	6.9	5.7	5.4	5.2	5.5	5.6	6.4	6.0	6.2
		MCDBR	8.6	10.4	9.6	9.2	7.3	7.0	6.3	6.7	7.5	8.8	8.4	9.7
	5% WB	MCWBR	5.8	7.0	7.1	7.2	5.8	5.3	5.2	5.5	5.8	6.6	6.5	6.7
(40) Clear-Sky Solar Irradiance	taub		0.334	0.331	0.324	0.306	0.281	0.268	0.280	0.299	0.315	0.324	0.321	0.327
	taud		2.492	2.506	2.511	2.479	2.363	2.260	2.303	2.373	2.394	2.403	2.459	2.485
	Ebn at Noon		966	929	869	772	650	567	631	759	860	925	973	982
	Edh at Noon		105	96	82	65	51	44	52	70	90	105	108	108
(44) All-Sky Solar Radiation	RadAvg		4.25	3.43	2.24	1.23	0.60	0.38	0.48	0.95	1.89	2.98	4.21	4.77
	RadStd		0.27	0.24	0.15	0.11	0.05	0.03	0.04	0.07	0.16	0.17	0.32	0.29

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page